

LECTURE 1: INTRODUCTION TO COMPUTING ETHICS

What is Ethics?

Ethics = the philosophy of morality. It deals with what is right and wrong behavior. Another useful definition: *"The rules or standards governing the conduct of the members of a profession."* (American Heritage Dictionary)

Societal Ethical Codes (examples across cultures)

- **Babylonia:** "An eye for an eye."
- **Hebrew law:** "Love thy neighbor as thyself."
- **Ancient India (1500 BC):** Honesty, charity, nonviolence, modesty.
- **Confucius (China):** "What you do not want done to yourself, do not do to others."
- **Christianity:** "Do unto others as you would have them do unto you."
- **Marcus Aurelius (Rome):** There is a universal moral law; moral relativism is rejected.

Ethical Theories (VERY IMPORTANT — often asked)

1. Relativism

- There is no absolute right or wrong.
- Morality depends on the norms of one's culture — an action is right or wrong based on the society in which it is practiced.

2. Universalism — splits into two:

- **Consequentialism:** Judge an action by its *consequences*. Does it favor the common good?
- **Deontology:** Judge the action *itself* as right or wrong based on rules, not consequences.

Why Ethics Matters in Computing

Computing professionals have knowledge, access, and the ability to **cause harm or do good** that most people lack. That's why they need ethical codes — like doctors, lawyers, and engineers.

Common Themes in Computer Society Ethical Codes

1. Dignity and worth of other people
2. Personal integrity and honesty
3. Responsibility for your work
4. Confidentiality of information
5. Public safety, health, and welfare
6. Participation in professional societies
7. Knowledge and access to technology = social power

Computer-related Offenses (3 categories)

1. Offenses that existed before computers but are **facilitated** by computers (e.g., fraud)
2. Offenses **against** computer hardware and software (e.g., hacking, viruses)
3. **Invasion of privacy**

Professional Organizations with Ethical Codes (memorize a few)

- **ACM** — Association for Computing Machinery
- **IEEE Computer Society**
- **BCS** — British Computer Society
- **AITP** — Association of IT Professionals
- **ICCP** — Institute for Certification of Computer Professionals
- **CIPS** — Canadian Information Processing Society

Case-Study Rule of Thumb

For every "Is X ethical?" question, think: *Did the person misuse access or privilege they were given?* If yes, it is unethical — even if the intent seemed harmless (e.g., a programmer "fixing" his own overdraft charge is still **unethical**, because he abused his professional access).

LECTURE 2 — BIOETHICS & ETHICAL DECISION-MAKING

What is Bioethics?

Bioethics is the interdisciplinary study of ethical, social, and legal issues arising from advances in biomedicine, biotechnology, and life sciences.

- **Bio** = life, **Ethos** = behavior/character.
- Covers medical ethics, research ethics, and environmental policy.

Three Meanings of "Ethics"

1. Synonym for "morally correct" / justified principles.
2. Particular beliefs or attitudes held by a specific group (e.g., engineering codes of ethics).
3. **An area of study** – understanding moral values, resolving moral issues, justifying judgments. *(This is the main sense used in the course.)*

Values vs Morals vs Ethics (common exam question)

Term	Meaning
Values	What is important and worthwhile (basis for moral codes)
Morals	Codes of conduct – values put into action
Ethics	Systematic, rational method to resolve dilemmas

Moral Dilemma

A situation where **two or more moral obligations, rights, or ideals come into conflict**. To resolve one, ask:

- Who are the stakeholders?
- What are the options?
- What justifications exist? Are they strong or weak?

Ethics vs Law (KEY DIAGRAM – often in paper)

Four possible combinations:

	Ethical	Unethical
Legal	Legal & Ethical (ideal)	Legal & Unethical (e.g., lying to a friend)
Illegal	Illegal & Ethical (e.g., civil disobedience)	Illegal & Unethical (e.g., theft)

Four Core Principles of Bioethics (MEMORIZE)

1. **Non-maleficence** – Do no harm
2. **Beneficence** – Do good
3. **Justice** – Be fair
4. **Autonomy** – Respect a rational person's right to make an informed, un-coerced decision

Plus two supporting principles:

- **Truthfulness**
- **Confidentiality** (e.g., doctor–patient)

Roles of Codes of Conduct (8 roles)

1. **Shared Standards** – common values across a diverse profession
2. **Positive Support to Act Ethically** – "I'm bound by my profession's code"
3. **Guidance on Obligations** – what engineers/professionals must do
4. **Motivation** to act ethically
5. **Education** – case studies and training
6. **Deterrence** from being unethical
7. **Personal Discipline** – formal basis for investigating misconduct
8. **Professional Image** – builds public trust, reduces outside regulation

Solving Ethical Problems by Consensus

Based on: reason, genuine debate, traditions, openness to criticism, protection of minorities.

"Ethical problems are NOT solved in isolation – and take time."

LECTURE 3 – COMPUTING ETHICS & PHILOSOPHY OF ETHICS

Rapid Pace of Change

From 1940s (first computer) → 1956 (1-ton 5MB hard disk) → 1991 (1 MHz space shuttle computer) → 2006 (1 TB pocket devices, 100 MHz car computers). **Impact:** Many everyday devices are now computerized – TVs, phones, watches, cameras, ovens, cars.

New Developments & Their Ethical Issues

- **Blogs** — amateur outlets that grew into news alternatives & PR tools.
- **Video sharing (YouTube, etc.)** — rise of amateur content + copyright infringement.
- **Kill switches** — malicious apps can be remotely disabled, but hackers could abuse this.
- **Social networking** — Facebook, MySpace, Classmates.com (1995). Raises questions about relationships, privacy, and physical health.
- **Crowd-funding** — funding via the internet.
- **E-commerce** — free stuff (email, games, news); Amazon grew from 1994 to \$8.5B in 10 years.
- **AI, Robotics & Motion sensors** — narrow specialized skills; personal assistants; airbag triggers.
- **Tools for Disabled People** — screen readers, speech recognition, prosthetics.

Key Issues

Unemployment, Crime, Loss of Privacy.

Themes

1. **Old problems in new context** — crime, copyright, gambling reappear online.
2. **Adapting to new technology** — law and behavior must catch up.
3. **Varied sources of solutions.**
4. **Perfection is a direction, not an option.**
5. **Trade-offs** — more security = less convenience.
6. Difference between **personal choice, business policy, and law.**

Types of Rights (often asked)

- **Negative rights (liberties)** — right to act *without interference* (life, liberty, pursuit of happiness).
- **Positive rights (claim-rights)** — obligation on others to *provide* something (e.g., a job, internet access).

Natural Rights

Respect for fundamental rights: life, liberty, property + increasing happiness.

Virtue Ethics (Aristotle)

"One lives a virtuous life by **doing** virtuous acts." Doing your work honestly, responsibly, ethically, and well IS a virtuous activity.

Rawls — Social Contract & Veil of Ignorance

- A system is just only if all citizens can actually use their freedoms.
- An action is **not ethical** if it leaves the least-advantaged people worse off.
- **Veil of Ignorance:** Imagine designing society without knowing your class, privileges, or personality — forces you to design fairly.

James Moor — Vacuum of Policies

Computers create **new ethical questions** with no existing rules. Moor says we need a **logical, conceptual framework** to formulate computer-ethics policy.

Invisibility Factor (KEY — often a 4–5 mark question)

Three types of "invisibility" in computing:

1. **Invisible abuse** — hidden computer operations used unethically.
2. **Invisible programming values** — programmer biases baked into code.
3. **Invisible complex calculation** — massive computations users can't verify.

John Barlow — Cyberspace

- Cyberspace is "a vast continent with no end to its dimensions."
- We need **new social contracts for the digital domain.**
- Concerns: privacy, freedom of expression, economic opportunity, property rights, culture.

Society, Morality, Ethics (quick recap)

- **Society** — an association of people organized under rules for the common good over time.
- **Morality** — rules of conduct; what people must and must not do.
- **Ethics** — the philosophical study of morality.

LECTURE 4 — STRUCTURE OF ORGANIZATION

Why Legal Structure Matters

The law recognizes *individuals* as responsible for their actions. Individuals can:

- Enter contracts
- Be sued
- Be tried for crimes
- Give evidence

For any business beyond the smallest, it's better to give the organization a **legal existence** through **incorporation** ("making into a body").

- Organizations seeking incorporation must convince the **Privy Council** their activities are in the public interest.
- **Statutory bodies** (e.g., District Health Authorities) are established by Act of Parliament.

Legal Forms of Business Organization

1. Sole Trader (unincorporated)

- An individual operating their own business.
- **No legal formalities** to start — you just begin trading.
- May need to register with Customs & Excise and Inland Revenue for tax, but *not* to exist as a sole trader.
- **Risk:** unlimited personal liability for all debts.

2. Partnership (unincorporated)

- Defined by the **Partnership Act 1890**: *"the relationship which subsists between persons carrying on a business in common with a view to profit."*
- Automatic — even without a formal agreement, two people doing business together are a partnership.
- **Risk:** partners are personally liable for all partnership debts, including those caused by another partner exceeding their authority.
- Mostly **out of favor** for modern business because of this risk.

3. Limited Company (incorporated) — the typical form for trading organizations.

Formal vs Informal Organization

- **Formal organization** — the official hierarchy, roles, rules, reporting lines.
- **Informal organization** — friendships and social networks that emerge naturally.
- Informal structures **arise from** formal ones; when people interact regularly they bond over shared interests. You cannot force rigid formality.

Case Study: Air France–KLM Merger (2004) — likely case-study question

Strategic Rationale ("1 + 1 > 2"):

- European consolidation to compete with global rivals and low-cost carriers.
- **Complementary networks** — Air France (Paris CDG hub) + KLM (Amsterdam Schiphol).
- Target: **€1.4 billion** in annual savings by 2010 through fleet modernization, joint purchasing, and the Flying Blue loyalty program.
- Dominated transatlantic routes.

Governance — "Holding Company Model":

- **Dual-brand strategy:** both airlines kept their names and hubs.
- Led by a **Group Executive Committee (GEC)** and a Board of Directors.
- First non-French CEO: Benjamin Smith (to bridge internal divides).
- French gov (28.6%) and Dutch gov (9.3%) both hold shares — creates political complexity.

Critical Challenges:

- **Cultural clash:** Dutch see French as hierarchical and aloof; French see Dutch as profit-focused and individualistic.
- **Strategic paralysis** — Dutch government's 2019 surprise share purchase caused friction.
- **Labor issues** — rigid French labor structures → strikes and high costs vs rivals like Ryanair.

Lesson: A successful merger needs financial logic AND cultural integration. Technical synergies alone don't guarantee success.

LECTURE 5 — FINANCE & ACCOUNTING

Core Definitions (LEARN CAREFULLY)

Revenue (a.k.a. turnover or gross income) The total income from selling goods/services in the business's primary operations.

Formula: Revenue = Price × Quantity Sold (BEFORE any expenses deducted). It's the **top-line** figure of the income statement.

Profit Financial gain when revenue **exceeds** expenses, costs, and taxes.

- Primary measure of company success.
- Determines ability to grow and reward investors.

Revenue vs Profit (one-liner): *Revenue = money coming in. Profit = what remains after all expenses are paid.*

Liquidation The formal process of closing a business. Assets are **sold** to pay off debts, creditors, and liabilities. Once everything is distributed, the company is **dissolved** and ceases to exist.

Financial Image (IMPORTANT — likely long question)

Financial Image = the overall perception/reputation of a person, company, or organization regarding its **financial strength, stability, and credibility**.

Key Aspects (memorize the 5):

1. **Financial Stability** – ability to manage expenses, debts, and income.
2. **Profitability** – how well the business generates profit from operations.
3. **Creditworthiness** – trust from banks, investors, suppliers when lending.
4. **Financial Transparency** – honest and clear reporting.
5. **Reputation with Investors & Customers** – confidence that attracts capital and customers.

Why a strong financial image matters:

- Attracts investors and funding
- Builds trust and credibility
- Improves business partnerships
- Supports long-term growth

Examples (pick one or two to cite):

- Global: Apple, Microsoft, Alphabet, Amazon, NVIDIA, Visa.
- Software globally: Microsoft, Oracle, Adobe, Salesforce, SAP.
- Pakistani software houses: **Systems Ltd, NetSol Technologies, 10Pearls, Arbisoft, Contour Software.**
- Pakistani companies: **OGDCL, HBL, Meezan Bank, Lucky Cement, PSO, State Life Insurance.**

Cost–Benefit Evaluation Techniques (VERY LIKELY CALCULATION QUESTION)

The four techniques:

1. Net Profit
2. Payback Period
3. Return on Investment (ROI)
4. Net Present Value (NPV)

1) Net Profit

$$NP = \text{Total Income} - \text{Total Cost}$$

- Very simple; add the cash flows column-by-column over the project.
- **Does NOT consider the time element** (money today vs money later).
- Of limited use on its own.

2) Payback Period (PBP) "How soon do we get our cash back?" – break-even point.

- Short payback periods are preferred.
- **Does NOT consider income/expenditure after break-even.**

If cash inflows are constant:

$$PBP = \text{Initial Investment} \div \text{Annual Cash Inflow}$$

Worked examples:

Example 1: Investment = \$100, annual inflow = \$20 → $PBP = 100 / 20 = 5 \text{ years}$

Example 2: Investment = Rs 25,00,000; annual inflow = Rs 5,00,000 for 8 years → $PBP = 2,500,000 / 500,000 = 5 \text{ years}$

Example 3 – Comparison:

	Machine A	Machine B
Initial investment	1,500,000	2,000,000
Annual cash inflow	500,000	550,000
PBP	3 years	3.6 years

Answer: Choose **Machine A** – shorter payback period.

If cash inflows are NOT constant → use the Cumulative Method: Add cash inflows year-by-year until the total equals or exceeds the initial investment.

Example: Initial investment = \$50,000

- If by end of year 3, cumulative = \$45,000 and year-4 inflow = \$10,000, then after year 4 cumulative = \$55,000 (exceeds investment).
- Fraction = $(50,000 - 45,000) / 10,000 = 5,000 / 10,000 = 0.5$
- **PBP = 3 + 0.5 = 3.5 years**

(Note: ROI and NPV were listed but not fully expanded in the lecture slides – be ready to define them if asked: $ROI = (\text{Net Profit} / \text{Investment}) \times 100\%$; NPV discounts future cash flows back to present value.)

LECTURE 6 — ANATOMY OF A SOFTWARE HOUSE

Company Structure (hierarchy)

Managing Director at the top, with the company split into five divisions:

1. **Technical**
2. **Sales & Marketing**
3. **Operations**
4. **Financial**
5. **Overseas**

Operations Director — key role

Responsible for **all revenue-earning operations**:

- Ensure all projects are completed satisfactorily.
- Maintain a satisfactory staff level.
- Match people to projects **optimally** — balancing staff utilization, individual learning wishes, overseas postings, short-term vs long-term goals. (*"Clashing structures"*)

Technical Director — responsibilities

- **Quality management**
- **Research and development**
- **Technical marketing** — arranging staff to give papers at conferences.
- **Technical training** (*not* project-management or presentation training — those belong to personnel/HR).

Management of Staff

A software house's **reputation depends on its staff quality**. Key tasks:

- **Recruit and retain** good staff.
- Encourage **loyalty**.
- Provide training (sometimes via other organizations).
- Avoid **cliques** (narrow exclusive groups).
- Run **staff appraisal** so that:
 1. Achievements and contributions are recorded.
 2. Employees know what is expected for promotion.
 3. Training and career-development plans are made.
 4. Employees know how the company views their performance.

Ongoing Problems in Software Houses

1. **Continuity of management** as staff move between projects.
2. **Using group expertise** (e.g., the company's database experts) when they're on different projects.
3. **Maintaining standards and loyalty** as the company grows.

Producing the Budget — who feeds in what

- **Sales staff** — how much additional business they could sell (if staff available).
- **Personnel dept** — ability to recruit/retain; salary-trend estimates.
- **Operations director** — achievable staff utilization rates.
- **Specific expansion plans** — e.g., new office.
- **Financial director** — ability to raise funds.

Staff classification in budget:

- Technical/revenue-earning vs non-revenue-earning staff.
- Different grades have different training needs.
- Senior staff often used on non-revenue tasks like sales support.
- Senior technical staff are specialized → longer to re-deploy.

Monitoring Financial Performance

- **Monthly** — compare income and expenditure against the budget.
- Significant deviations → corrective action taken.
- **Project costing** — costs and revenue of each project tracked monthly, profit calculated as a percentage of total revenue.

Two Key Sales Reports

- **Confirmed Sales Report** — for each grade, number of staff committed to contracts in each of the next 12 months + expected revenue.
- **Sales Prospects Report** — for each prospect, its potential value, likelihood, and start date.

Long-term Planning (2 goals)

1. Identify appropriate **long-term goals**.
2. Identify and formulate plans to **overcome structural problems**.

Expansion Plans depend on:

- Ability to **recruit and retain** suitable staff.
- Ability to **raise funding**.
- Ability to **run a larger, often multinational** organization.

Company Image (software house) — key factors

Built through professionalism, quality of work, communication, branding, and reputation.

- Deliver high-quality solutions **on time and within budget**.
- **Transparency** and reliable post-deployment support.
- Strong **brand identity** — professional website, social media, certifications, portfolio.
- **Internal culture** of teamwork, learning, ethics, and innovation reflects outward.

Finance — common problem

Under-capitalization — not enough money invested in the company. Must be funded from:

- **Profits**
- **Loans** secured against directors' personal property
- **New equity capital**

Conclusion (think of this as the essay skeleton)

A software house needs these departments working together: project management, software development, QA, UI/UX, DevOps, marketing, client support. Success requires:

- Clear requirements analysis
- Structured methodologies (Agile / Waterfall)
- Proper documentation
- Effective leadership
- Continuous improvement and teamwork

LECTURE 7 — INTELLECTUAL PROPERTY (IP)

What is Property?

Article 260 of the Constitution of Pakistan (1973) says:

"Property includes any right, title or interest in property, moveable or immoveable, and any means and instruments of production."

Intellectual Process

Whatever a person creates by **applying the mind** is called an intellectual process. It is the source of creativity and invention. If not rewarded, people stop creating — so IP rights **encourage innovation**.

What is Intellectual Property?

IP = a category of property that includes intangible creations of the human intellect. Main types: **copyrights, patents, trademarks, trade secrets.**

Two Broad Forms of IP

1. **Literary & Artistic Works** — books, paintings, music, plays, films, TV/radio programs, performances. → Protected by **COPYRIGHT**.
2. **Industrial Property** — physical matter produced from an idea for commercial use. → Protected by **Patents, Trademarks, Industrial Designs, Trade Secrets, Layout-Designs, Geographical Indications.**

The 7 Categories of IP (modern view)

1. **Copyright** — literary, artistic, scientific works (books, magazines, articles).
2. **Related rights** — performances, broadcasts, concerts, shows.
3. **Trademarks** — marks, service marks, commercial names, logos (e.g., champagne).
4. **Geographical indications** — words showing the origin of goods/services.
5. **Patents** — inventions (new engine, floor cleaner, etc.).
6. **Industrial designs** — shape of a bottle, machine, luxury car.
7. **Unfair competition** — protection against false claims or imitation that deceives customers.

What are Intellectual Property Rights?

Rights given to authors/creators of new and original literary and artistic works (books, articles, paintings, music, sculpture, films, computer programs).

- **Compensate** creators for their effort and investment.
- Granted for a **limited time**; afterwards the public can use the work freely.

Necessity of IP Rights Protection (4 reasons — MEMORIZE)

1. **Encouragement to creativity by ensuring reward** — without reward, no one bears the effort of creation.
2. **Innovations in technology** — no one invests in R&D without a return; IP gives the incentive.
3. **Protection of users and consumers** — pirates/imitators sell fake products that harm buyers and damage real producers. Trademarks let consumers make informed choices safely.
4. **Transfer of technology** — proper IP protection enables tech transfer (through FDI, franchising, licensing, joint ventures) from developed to less-developed nations, generating jobs and growth.

WIPO (World Intellectual Property Organization)

- Established by the **WIPO Convention, 1967**.
- **Specialized agency of the United Nations**.
- Promotes IP protection **worldwide**.
- **Headquarters: Geneva, Switzerland**.
- Activities: mass-awareness campaigns, World IP Day, collaborating with national institutions, seminars, workshops, training.

IP Establishments in Pakistan (key law + ministry)

IP laws in Pakistan are administered by **three ministries**:

1. **Ministry of Education** — Copyright Ordinance, **1962**.
2. **Ministry of Industries and Production**:
 - Registered Designs Ordinance, **2000**
 - Registered Layout-Designs of Integrated Circuits Ordinance, **2000**
 - Patents Ordinance, **2000**
3. **Ministry of Commerce**:
 - Trade Marks Ordinance, **2001**
 - Merchandise Marks Act, **1889**

Offices (all in Karachi):

- **Central Copyright Office** — under Ministry of Education
- **Trade Marks Registry** — under Ministry of Commerce
- **Patent Office** — under Ministry of Industries and Production

Copyright Ordinance, 1962

Copyright = the exclusive right to do (or authorize) any of the following with a literary, dramatic, musical, artistic, or cinematographic work:

- Reproduce the work in any material form
- Publish the work
- Perform the work in public
- Use it in a cinematographic work
- Show it on television

Works in which copyright subsists:

- Original literary, dramatic, musical, and artistic works
- Cinematographic works
- Records (sound recordings)

Broadcasting organizations' rights: authorize re-broadcast and fixation of broadcasts — **subsists 25 years**.

Copyright does NOT cover: ideas, facts, recipes, works lacking originality (e.g., the phone book), names/titles/short phrases.

Registered Designs Ordinance, 2000

- A design can be registered by its proprietor for an article or set of articles.
- Must be **new or original**.
- Cannot be registered if the same design has already been registered in Pakistan or published anywhere in the world, or differs only in immaterial details.
- **Penalty:** imprisonment up to **2 years** or fine up to **Rs 20,000** (or both).

Patents Ordinance, 2000

An invention is patentable if it is:

1. **New**
2. Involves an **inventive step**
3. **Capable of industrial application**

NOT considered inventions (cannot be patented):

- A discovery, scientific theory, or mathematical method
- A literary, dramatic, or musical work / purely aesthetic creation
- A scheme, rule, or method for performing a mental act, playing a game, or doing business

Patents are territorial — there is **no global patent**. Granting or refusing a patent in one country does not affect others.

Famous example: A 9-year-old wheelchair-bound boy from Jaipur, Hridayeshwar Singh Bhati, invented and patented a 6-player circular chess game.

Trade Marks Ordinance, 2001

A trademark may be registered for **goods, services, or both**, following the international classification.

A **trademark can be**: a name, logotype, symbol, slogan, shape, or color.

Infringement — Do's and Don'ts

- **Create yourself** rather than using others' creations.
- Do **not** use a competitor's mark in a way that harms them unfairly.
- Do **not** make comparisons that cause customer confusion.

Real example: Apple sued HTC, claiming 20 Apple patents relating to the iPhone were being infringed. Steve Jobs: *"Competitors should create their own original technology, not steal ours."*

Practical Case Study (often tested — Lumen/Mastery Business Law)

A large company discovers a small company infringing its patent. It writes asking them to stop. The small company denies it. Correspondence is lost for 8 years due to personnel changes. The large company then sues.

Likely result: Suit may be barred by **laches / delay** — the large company failed to act promptly, and equity does not help those who sleep on their rights. Identify → delay in asserting rights; Reason → the small company was not put on clear continued notice and may have invested; Result → claim weakened or dismissed (or damages limited to the period within the statute of limitations).

LECTURE 7b — APPLIED ETHICS & IP CASE STUDIES

Healthcare Software Case (CRITICAL — almost certainly on the paper)

Scenario: A QA team finds occasional erroneous reports in a healthcare app. Fixing takes 5 weeks. The manager says: *"Ship now, fix in updates — missing the deadline hurts our reputation."*

Q: Is it ethical to release software with known defects? No. Three reasons:

1. **Duty of care** — healthcare software directly affects patient lives.
2. **Professional integrity** — engineers/QA must deliver reliable products, especially in critical domains.
3. **Deceptive practice** — knowingly shipping a defective product misleads client and end-users.

Q: What are the risks to patients?

- Misdiagnosis or delayed treatment
- Wrong medications / missed interventions → health deterioration
- Loss of trust in the software and healthcare system
- Legal consequences → malpractice claims and liability

Q: Should the team follow the manager or report the issue? The team should **prioritize patient safety over deadlines**. Steps:

1. **Document** the defect and its potential impact.
2. **Escalate responsibly** to higher management, the client, or a compliance officer — emphasizing safety.
3. **Refuse unsafe release** — violating ethical and legal standards is not acceptable.
4. **Suggest alternatives** — release a limited/non-critical version, or delay until fixed.

Key principle: In healthcare software, **patient safety always takes precedence over deadlines or reputation**.

Contract and Delivery for Software (typical LMS contract contents)

A proper software-delivery contract must include:

1. **Software Product** — all modules (user management, course management, content management, assessments & grading, communication tools, reporting & analytics, integration).
2. **Documentation** — user manuals, admin guides, technical documentation (architecture, APIs, DB schema, code docs).
3. **Testing & QA** — test plans, test cases, unit/integration/UAT results, bug-tracking logs.
4. **Deployment & Training** — full deployment on chosen platform, training sessions, post-deployment support.
5. **Source Code & Ownership** — all code and assets transferred on acceptance; IP rights defined in a separate clause.

Actions to Establish & Promote an Ethical Culture (6 points)

1. **Clear policies and a Code of Ethics** covering fairness, workload, transparency, and behavior — discourage uncompensated overwork.
2. **Promote work-life balance** — reasonable hours, no mandatory overtime, flexibility.
3. **Transparent communication and feedback** — regular meetings, anonymous ethics hotline.
4. **Recognition and rewards** — fair pay, performance incentives, reward collaboration *and* ethics.
5. **Leadership by example** — managers must model empathy, fairness, and integrity.
6. **Training and awareness** — regular ethics workshops.

Long-Term Career Planning — Factors to Consider (7 points)

1. **Employee skills and aspirations** — understand goals via appraisals and 1-on-1s.
2. **Growth opportunities** — clear technical, managerial, specialist paths; mentorship.
3. **Training and development** — courses, certifications, cross-functional projects.
4. **Succession planning** — prepare employees for critical future roles.
5. **Work-life balance and retention** — avoid burnout.
6. **Performance metrics and feedback** — link evaluation to promotions/raises.

7. **Alignment with organizational goals** – mutual benefit for employee and company.

Types of IP – Quick Reference (for a scenario question)

Given a scenario like "Company DataNova has a unique algorithm + UI":

- **Trade Secret** – the algorithm (confidential, gives competitive advantage).
- **Copyright** – UI design, source code, documentation (original expression).
- **Patent (potential)** – if the algorithm is **novel, non-obvious, and useful**, it can be patented (exclusive rights, usually 20 years).
- **Trade Dress** – distinctive UI that identifies the source of the software.

Strategies to Protect Trade Secrets (4 strategies)

1. **Non-Disclosure Agreements (NDAs)** for all employees, contractors, collaborators.
2. **Access control and confidentiality policies** – limit access, secure authentication.
3. **Employment agreements and exit protocols** – restrict ex-employees from sharing.
4. **Internal documentation** – record the steps taken to maintain secrecy (crucial for legal protection).

Preventive Measures

1. **File a patent (or provisional patent)** – prevents competitors from using the algorithm even if an employee leaves. Patent typically lasts **20 years** and attracts investors.
2. **Strong legal agreements** – NDAs, non-compete clauses (where enforceable), clear IP ownership.

FINAL EXAM CHECKLIST – WHAT TO REVISE IN THE LAST HOUR

1. **4 Ethical Theories** – Relativism, Universalism (Consequentialism, Deontology), plus Virtue Ethics and Rawls' Veil of Ignorance.
2. **4 Bioethics Principles** – Non-maleficence, Beneficence, Justice, Autonomy (+ Truthfulness, Confidentiality).
3. **Ethics–Law 2x2 matrix** – legal/illegal x ethical/unethical.
4. **Invisibility Factor (James Moor)** – abuse, programming values, complex calculation.
5. **3 Legal forms**: Sole Trader, Partnership (Act 1890), Limited Company; incorporation.
6. **Revenue vs Profit vs Liquidation** – definitions.
7. **Payback Period** – both constant formula and cumulative method. Practice 2 examples.
8. **Software house roles** – Operations Director, Technical Director; budget inputs.
9. **7 categories of IP + 4 reasons why IP protection is necessary.**
10. **WIPO** – 1967, UN agency, Geneva.
11. **Pakistan IP laws** – Copyright Ordinance 1962, Designs/Layout/Patents Ordinances 2000, Trade Marks Ordinance 2001.
12. **Patent requirements** – new, inventive step, industrial application; what is NOT an invention.
13. **Healthcare Software Case** – patient safety > deadlines; escalate, document, refuse, suggest alternative.
14. **Trade secret protection** – NDAs, access control, exit protocols, documentation.

Writing Tips for Full Marks

- **Start every definition with the textbook phrasing** – examiners reward exact wording.
- **Use bullet points** for multi-part answers (readability = marks).
- **Always give one example** per concept – shows you understood it.
- **Case studies need 4 parts**: Issue → Stakeholders → Ethical principle → Recommended action.
- **Finance questions**: write the formula first, then plug values, then state the answer clearly with units. Draw the table if comparing options.
- **Don't leave any question blank** – even a partial answer with the right definition earns marks.

Good luck – you've got this! 🍀